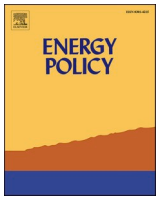
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Renewable energy trade destruction and deflection: Effects of antidumping and countervailing measures on Chinese solar photovoltaic exports

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ABSTRACT

In the context of the worldwide shift toward sustainable energy and escalating trade barriers, the domestic solar energy sector encounters ongoing challenges from antidumping and countervailing measures. This research explores the adverse impact of antidumping and countervailing policies (ADCV) on Chinese Solar Photovoltaic Exports (CSPE), as well as how CSPE shift to third-party markets in response to these trade protection measures, utilizing a panel dataset from 2007 to 2020, with analysis conducted using a staggered DID model. The results show that (1) ADCV have trade-destruction effects and substantially reduce CSPE. (2) The trade deflection effects indicates that when CSPE are restricted by ADCV, they are redirected to third-party nations with strong PV demand, infrastructure, and energy transitions. Additionally, some production capacity shifts to domestic sales, mitigating the negative impact of trade disruptions. (3) The heterogeneity analysis reveals that ADCV affect China's PV exports differently based on varying policy enforcement, energy investment levels, and energy dependence across countries. The findings of this paper provide valuable insights for China to deal with renewable energy trade barriers and optimize the overseas market layout of the PV industry.

1. Introduction

In recent years, as the global climate change problem has intensified, the need for energy transition has become more urgent, and countries have increased their investment and policy support for renewable energy. The growth of the renewable energy sector is a key focus in the economic strategies of numerous nations and is considered a crucial area for advancing the energy transition (Su et al., 2023; Carley, 2022; Bui and Tseng, 2021; Xu and Lin, 2025). Photovoltaic (PV), an essential form of renewable energy, has received widespread attention and is rapidly developing. With its abundant manufacturing resources, advanced technology, and large-scale production capacity, China has quickly become the leading global producer and exporter of photovoltaic products (Zhu et al., 2021a, 2021b; Gao and Rai, 2019). The pace of globalization in the Chinese PV sector is reflected not only in the rapid expansion of its market share but also in its promotion of the optimization of the global PV industry chain and technological innovation. By continuously improving product quality, reducing production costs, and upgrading technology, Chinese PV sector has not only strengthened its competitive edge in the international market but also made significant contributions to the growth of the global photovoltaic sector and the progress of the carbon neutrality process (Lin and Liu, 2024; Jackson et al., 2021; Yao and Cai, 2019; Ogura, 2020).

However, as the competitiveness of Chinese photovoltaic products (CPVP) continues to rise in global markets (Luo et al., 2021; Binz and Anadon, 2018), some countries have begun to implement ADCV on CPVP out of concern for the protection of their PV industries (McCarthy et al., 2016; Wang et al., 2020; Hughes and Meckling, 2017). These policies aim to counter the low-price competition of Chinese PV products, protect the domestic industry from shocks, and maintain domestic employment and economic stability (Chen et al., 2017; Ding et al., 2019; Voituriez and Wang, 2015). The antidumping and countervailing measures in major markets like the EU and the US have disrupted the growth of the global photovoltaic sector and notably affected CSPE. Therefore, Driven by a range of factor, such as the increase of geopolitical risks in the world, the rise of trade protectionism in China, and the attempts of various countries to build their renewable energy industry chain, the risk of China's photovoltaic industry encountering ADCV in Europe and the US will always exist. It is crucial for China's photovoltaic industry to

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Nomenclature abbreviations

ADCV	antidumping and countervailing
CSPE	Chinese Solar Photovoltaic Exports
CPVP	Chinese photovoltaic products
Units	
GW	Gigawatt
TWh	Terawatt-hour
EJ	Exajoule

effectively deal with green trade barriers and diversify its production capacity overseas. Layout has become a crucial issue that needs to be solved.

Existing literature on China's solar photovoltaic exports (CSPE) has provided valuable insights into the influence of both domestic and international factors. On the domestic front, scholars have highlighted the role of policy incentives, such as subsidies, feed-in tariffs, and national development strategies, in fostering the growth of China's PV industry (Lin and Luan, 2020; Wang and Fan, 2021; Wang et al., 2016; Liu et al., 2016; Lin et al., 2023). On the international front, increasing attention has been paid to the influence of global market conditions and trade policies, particularly trade protectionism, on China's renewable energy sector (Che et al., 2022; Nwanekezie et al., 2022; Li et al., 2021; Li and Fang, 2024; Hajdukiewicz and Pera, 2020; Hughes and Meckling, 2017).

Among these, anti-dumping and countervailing (ADCV) measures have emerged as critical external constraints for China's export-oriented PV industry. However, three important gaps remain. First, while many studies analyze the overall impact of ADCV measures on Chinese industries (e.g., Cheng et al., 2021), relatively few focus specifically on the PV sector, even though it represents a flagship export industry within China's broader green development strategy. Given the sector's rapid globalization and strategic importance, it is essential to understand the specific implications of ADCV policies and the corresponding responses of China's solar photovoltaic exports (CSPE). Second, existing research primarily concentrates on bilateral trade disputes, such as those between the United States and China or the European Union and China, without providing a comprehensive view of the broader global landscape of ADCV actions targeting China's PV exports (Huang et al., 2024; Nguyen et al., 2019; Lee et al., 2013; Felbermayr and Sandkamp, 2020; Jabbour et al., 2019). This narrow focus limits our understanding of the international dynamics shaping China's renewable energy trade. Third, although the literature has extensively discussed the trade-disrupting effects of protectionist policies (Li and Li, 2022; Li et al., 2014), much less attention has been given to trade deflection, which refers to the rerouting of exports to alternative markets. Given China's extensive global trade network, examining how and where CSPE are redirected is crucial for generating theoretical insights and informing policy formulation (Yuan et al., 2023; Chen et al., 2022; Timilsina et al., 2022).

To fill these research gaps, this paper systematically examines the effects of ADCV policies on CSPE using a staggered difference-in-differences approach. We investigate both the trade-disruption and trade-deflection effects of ADCV actions imposed by multiple countries, as well as China's strategic responses through international market diversification and domestic PV infrastructure development. The findings offer new empirical evidence for understanding the complex trade-offs between trade protectionism and renewable energy development, and provide policy-relevant insights for China's ongoing efforts to strengthen its position in the global PV value chain.

This paper makes three key contributions to the literature.

First, most empirical studies have focused on industry-wide effects of anti-dumping and countervailing measures, with few targeting specific renewable energy sectors such as photovoltaics, and most relying on qualitative methods (Cheng et al., 2021; Lin and Luan, 2020; Wang and Fan, 2021). This paper fills this gap by using a staggered differences-in-differences approach to quantitatively examine the trade-destructive and trade-deflection effects of ADCV measures on China's solar photovoltaic exports (CSPE).

Second, existing literature has primarily emphasized the "trade disruption" impact of ADCV, while relatively little attention has been paid to trade deflection and the underlying mechanisms behind the redirection of exports (Li and Li, 2022; Yuan et al., 2023; Chen et al., 2022). This paper contributes by analyzing not only the disruption effects but also identifying the country characteristics that attract China's PV export deflection under ADCV pressures, thus enriching the understanding of trade deflection dynamics.

Third, existing research lacks a systematic examination of how China strategically responds to renewable energy trade barriers through both external and internal channels. Most studies merely identify the negative impacts of ADCV measures, without exploring subsequent adjustment strategies, particularly those related to export market reorientation and domestic demand absorption (Timilsina et al., 2022; Li and Fang, 2024). This paper fills this gap by offering concrete policy-oriented insights, analyzing how China diversifies its export destinations while simultaneously promoting the development of its domestic photovoltaic power generation capacity, thereby enhancing the resilience and international competitiveness of its PV industry under global trade restrictions.

The structure of this paper is outlined as follows: Section 2 discusses background of the implementation of ADCV. Section 3 outlines the data, variables, and model used in the analysis. Section 4 presents the empirical findings and offers an in-depth analysis. Section 5 offers additional insights. Finally, Section 6 concludes with a summary of the key results and provides specific policy recommendations.

2. Background of the implementation of ADCV

2.1. Implementation of ADCV

Globally, China's solar energy industry has been heavily targeted by ADCV implemented by several countries due to its rapid development and significant market share (Lee et al., 2017; Sandkamp, 2020). These investigations have profoundly impacted CSPE. Table 1 gives the specific details of countries' ADCV against Chinese PV. The U.S. ADCV on Chinese PV products began in 2011. In November that year, the US Department of Commerce initiated antidumping and countervailing investigations. In March 2012, the US formally imposed countervailing duties on Chinese PV products, ranging from 14.78 % to 15.97 %. Subsequently, in May 2012, the US Department of Commerce ruled that Chinese PV products had dumping behaviors, with dumping margins ranging from 18.32 % to 249.96 %, and began to impose antidumping duties. This series of measures notably increased the cost of CPVP entering the US market, forcing Chinese companies to adjust their export strategies (Zhu et al., 2021a, 2021b). The EU initiated an anti-dumping investigation into Chinese PV products in September 2012, followed by a countervailing investigation in November 2012. In June 2013, the European Commission ruled that Chinese PV products were dumped and initially chose to implement a temporary tariff of 11.8 %, after which this rate was raised to 47.6 %. At the same time, the EU determined that Chinese PV products were receiving unfair government subsidies, with countervailing duty rates ranging from approximately 6.5 %–11.5 %. Although in August 2013, The EU and China agreed on a price commitment, establishing a minimum export price for shipments to the EU, in 2017, the EU again decided to extend the double-reverse measures until September 2018, continuing to put pressure on Chinese PV firms (Ball et al., 2017). In July 2017, India initiated an anti-dumping investigation into Chinese PV products. After a one-year investigation, India's Anti-Dumping Authority ruled in 2018 that there was dumping of CPVP, and imposed antidumping duties at rates ranging from $0.15 to $0.45 per watt. This measure further limited the competitive strength of

Table 1
Implementation of anti-dumping and countervailing policies.

Country	Formal Imposition Date	Countervailing Duty Rate	Anti-Dumping Duty Rate	Policy Details
USA	March 2012 (CV), May 2012 (AD)	14.78 %–15.97 %	18.32 %–249.96 %	Initiated ADCV investigation in November 2011. Imposed CV in March 2012 and AD in May 2012.
EU	June 2013 (provisional AD), August 2013 (price undertaking agreement), 2017 (extension)	6.5 %–11.5 %	11.8 % (provisional) increased to 47.6 %	Initiated AD investigation in September 2012 and CV investigation in November 2012. Imposed provisional AD in June 2013, followed by a price undertaking agreement in August 2013, and extended measures in 2017 until September 2018.
India	2018	None	$0.15 - $0.45 per watt	Imposed AD duties in 2018, further restricting the competitiveness of CPVP in the Indian market.
Canada	June 2015	1 %–77 %	9 %–286 %	Imposed AD and CV duties in June 2015, significantly increasing the difficulty of exporting Chinese PV products to Canada.
Turkey	2017	Varies by company	$20 per kilogram	Imposed AD and CV duties in 2017, aiming to protect domestic PV industry and limit low-priced Chinese PV imports.

CPVP in the Indian market. Canada commenced an ADCV investigation into CPVP in December 2014. in June 2015, the Canada Border Services Agency ruled that Chinese PV products were dumped and unfairly subsidized and imposed anti-dumping duties and countervailing duties at rates ranging from 9 % to 286 % and 1 %–77 %, respectively. This policy significantly increased the difficulty of exporting Chinese PV products to Canada. Turkey initiated ADCV investigations into CPVP in 2016. After a period of investigation, Turkey decided to impose ADCV on CPVP in 2017. Specifically, the anti-dumping duty rate is $20 per kilogram, while the countervailing duty rate varies according to companies. This policy aims to protect Turkey's local PV industry and limit the import of low-priced Chinese PV products.

The main reason for these countries to implement the Dual Anti-Reflection Policy (DARP) on Chinese PV products is that Chinese PV products have a competitive advantage in the global market with lower prices, which leads to the local PV industry in the importing countries to face severe market competition pressure and economic losses. By implementing the Dual Reverse Policy, these countries hope to limit the low-price competition of Chinese PV products, protect the growth of the domestic PV industry, and ensure a fair market competition environment (Kuang and Xiang, 2019). With the implementation of the double-reverse policy, China's PV industry is forced to adjust its export strategy, accelerate the pace of market diversification, and make industrial chain adjustments to cope with the ever-changing international trade environment (Liu and Shi, 2019; Lu, 2010).

2.2. China's photovoltaic exports under anti-dumping and countervailing policies

Since Europe, the United States, and other countries implemented ADCV against CPVP in succession in 2012, the export pattern of CPVP has experienced profound changes. With its status as the foremost producer of photovoltaic products globally, China plays a crucial role in global PV trade and consumption. However, the low prices of Chinese PV products have brought severe market competition pressure and economic losses to the local PV industry in importing countries. As a result, Europe, the United States, and other countries have implemented double-reverse policies on Chinese PV products to protect their domestic industries.

As shown in Fig. 1, this paper portrays in detail the changes in the trade volume of CPVP exported to each of the countries in Europe and the United States that implemented ADCV. In this paper, five countries, namely, the United States, Germany, Canada, India, and Turkey, are selected to observe the changes in the trade volume of CPVP while implementing ADCV. The U.S. has implemented ADCV on CPVP since 2012, which led to a significant decline in the trade volume exported to the U.S. after 2012. It kept it at a low level in the following years. Germany started implementing ADCV in 2012, leading to a significant decrease in exports of CPVP to Germany after 2012 and a rapid decline in trade volume from 2013 onwards, with overall trade volume well below the level before the policy was implemented. Canada began imposing ADCV on CPVP in 2015, significantly decreasing CSPE to Canada after 2015. India implemented ADCV on CPVP in 2018, and as can be seen from the chart, India's imports of Chinese PV products increased significantly in 2017 but declined rapidly after the implementation of the policy in 2018 and gradually declined over the following three years. Turkey started to implement ADCV in 2017, and the export value of CPVP declined rapidly after the implementation of the policy, especially between 2017 and 2018, when the trade volume decreased significantly. In summary, the import volume of Chinese PV products from each country implementing ADCV decreased significantly during the implementation of the policy, which preliminarily indicates that ADCV has had a significant inhibitory effect on the export of CPVP.

3. Research design

3.1. Data

This study utilizes panel data on CSPE from 2007 to 2020 to evaluate the impact of ADCV imposed by the US, EU, and other countries. The period from 2007 to 2020 is selected because it marks a phase of rapid growth in China's PV industry, during which ADCV policies were progressively implemented, significantly influencing China's PV export performance. The data on CSPE were obtained from UN Comtrade. The PV trade data (2007–2020) from UN Comtrade, the most extensive and authoritative database on international merchandise trade, includes contributions from over 200 countries and regions, covering 99 % of global trade. Similar to the sample selection process in previous studies, we extracted PV trade data between China and other countries from the database using HS code 854140(Lin and Liu, 2024; Monkelbaan, 2014; Algieri et al., 2011).

In addition to the PV trade data, other control variables used in this study were sourced from reputable global databases, such as the World Bank database, which provides comprehensive economic indicators, and the Energy Institute's Statistical Review of World Energy (EISRWE),

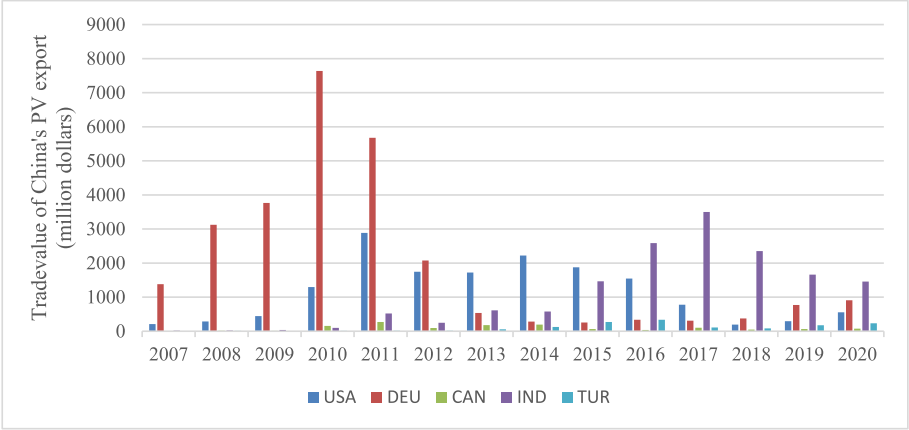


Fig. 1. CPVP exported to the ADCV countries.

which offers valuable data on global energy production, consumption, and trends. These sources helped ensure the robustness of our analysis by incorporating relevant macroeconomic and energy-specific variables. Furthermore, data related to the global construction of photovoltaic (PV) power plants were manually collected from the "Global Energy Monitor," a platform dedicated to tracking global energy infrastructure developments. This information, which includes data on new PV installations, was carefully organized and incorporated into our dataset to analyze the impact of both domestic and international PV power plant construction on CSPE.

3.2. Variables

3.2.1. Dependent variable

Consistent with previous studies, this paper measures CSPE using the trade volume of PV exports from China to other countries, while, to control the heteroskedasticity of the data and to improve the robustness of the model, we take the logarithmic value of the export volume to

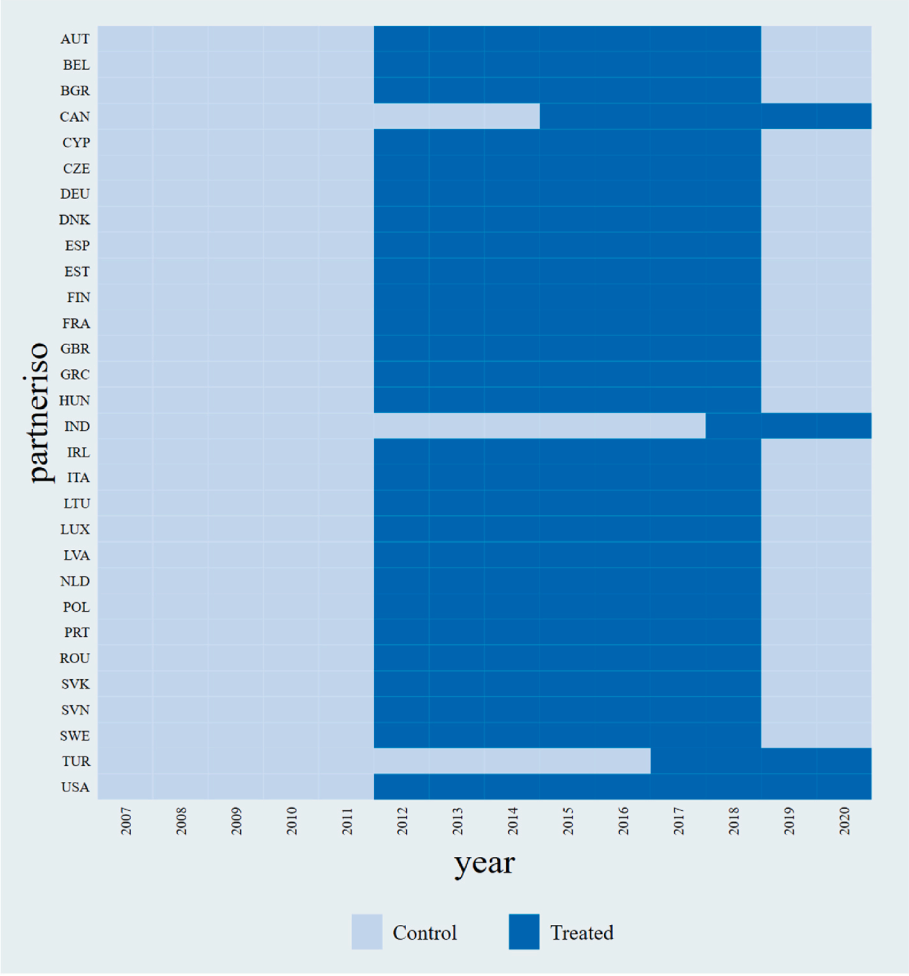


Fig. 2. Under control and under treated.

measure CSPE.

3.2.2. Independent variable

To study the impact of antidumping and countervailing policies (ADCV), this paper uses the interaction of time and individual dummy variables as the core explanatory variable to measure the effect of ADCV implementation. Fig. 2 presents the periods during which each of the ADCV countries in the treatment group imposed measures on China.

3.2.3. Control variables

Since the study in this paper involves bilateral flows of trade, the control variables are selected with reference to the explanatory variables commonly used in trade gravity models(Baier and Bergstrand, 2007). Following the approaches of Khan et al. (2024) and Lin and Liu(2024), the control variables used in this study include (1) Measured by the logarithm of real GDP, reflecting a country's overall economic size. A higher GDP typically correlates with greater demand for imported photovoltaic (PV) products. (2) Population size (lnpop): Represented by the logarithm of total population, as larger populations can drive higher demand for PV products due to increased energy consumption. (3) Value added in industry (lnind): The natural logarithm of industrial output, indicating a country's industrial base. A strong industrial sector often requires more PV products for energy supply and sustainability goals. (4) Urbanization level (urb): Measured by the proportion of the urban population, as urbanized areas tend to have higher energy needs and greater adoption of renewable energy technologies, including PV. These variables help control for differences in economic conditions and ensure a robust analysis of bilateral trade flows.

3.2.4. Mechanism variables

Following the imposition of antidumping and countervailing policies, China's PV exports not only declined in ADCV markets but also showed signs of redirection to third-country markets. To better investigate the underlying mechanisms of trade deflection and identify the characteristics of countries to which exports are redirected, this paper incorporates several mechanism variables reflecting PV market demand, infrastructure development, and renewable energy progress in importing countries. These variables help provide a more nuanced understanding of how trade policies influence the reallocation of China's PV exports across global markets.

(1) PV market demand: Market demand is a fundamental driver of trade flows. Countries with higher levels of solar power consumption or generation are more likely to import photovoltaic products to meet domestic energy needs and support their decarbonization goals (IEA, 2024; IRENA, 2023). Therefore, this paper includes three indicators to measure PV market demand: solar_ej (solar energy consumption in exajoules), solar_twh (solar electricity generation in terawatt-hours), and solar_twh_net (net solar electricity generation after accounting for transmission and distribution losses). Together, these variables reflect the overall scale and intensity of solar energy use in each importing country, which may significantly influence their attractiveness as alternative destinations for Chinese PV exports.

(2) PV infrastructure construction: The capacity and expansion of PV infrastructure directly shape a country's ability to absorb imported solar equipment. Countries with a greater number of PV power plants typically exhibit active deployment of utility-scale solar projects, which serve as key drivers of international PV trade (IEA-PVPS, 2024). To capture this aspect, this study includes the variable pvlplant, which represents the number of large-scale PV power plants (with a capacity of at least 20 MW) in each country. This variable functions as a proxy for both the scale of PV-related construction and the associated import potential of solar components.

(3) Renewable energy development: Countries that prioritize renewable energy development and maintain a high proportion of renewables in their electricity mix are more likely to increase demand for photovoltaic imports, thereby shaping the redirection of China's PV exports toward these alternative markets (REN21, 2024; UNEP, 2024). To measure this dimension, five variables are included: renewables_ej (total renewable energy consumption in exajoules), ren_power_ej (renewable power consumption in exajoules), ren_power_twh (renewable power generation in terawatt-hours),ren_power_twh_net (net renewable power generation), and electbyfuel_ren_power (the share of electricity generated from renewable sources). These indicators reflect the depth, scale, and pace of renewable energy transition in each importing country. Descriptive statistics for all variables are shown in Table 2.

3.3. Model specification

The ADCV imposed by Europe, the United States, and other countries on CSPE serve as exogenous shock. Given the varying timelines in which these countries implemented their policies against China, this paper employs a staggered DID approach to evaluate the effects of these measures on CSPE. This study examines two key dimensions of difference. First, from a temporal perspective, ADCV are divided into treatment and control periods, enabling a comparison of the impacts before and after policy implementation. Second, from the cross-sectional dimension, countries importing PV products from China are categorized into treatment and control groups based on whether or not they have enforced ADCV. It allows for a comparative analysis of the policy's effect on China's PV exports. By controlling for other factors, the staggered DID method enables assessing whether a significant difference exists in China's PV exports to countries that have enacted these policies versus those that have not before and after the policy enforcement. This method effectively quantifies the influence of ADCV measures on CSPE. The following regression model is constructed to test whether these policies have a suppressive effect on CSPE.

$$CSPE_{it} = \beta_0 + \beta_1 ADCV_{it} + \sum \beta_i control_{it} + \lambda_i + \delta_t + \epsilon_{it} \quad (1)$$

4. Empirical results and discussion

4.1. Baseline regression

Table 3 displays the results of the baseline regression model and explores the effect of ADCV on CSPE. In column (1), where no control variables are included, the regression coefficient between the ADCV and CSPE is −1.7587. Column (2) introduces economic scale (lngdp) as a control variable, yielding a coefficient of −1.6453 for the ADCV, which is also significant at the 1 % level. Adding total population (lnpop) in column (3) results in a coefficient of −1.5710. Industrial value-added (lnind) is added as a control in column (4), and the coefficient for ADCV becomes −1.5488. Finally, column (5) incorporates urbanization level (urb) as a control, with the ADCV coefficient adjusting slightly to −1.5344, remaining significant at the 1 % level. Across all specifications, the ADCV coefficient consistently shows a significant adverse effect on CSPE, demonstrating the robustness of the results despite the step-by-step addition of control variables.

Therefore, we find that the ADCV policies of the EU, the U.S., and other countries have a trade-destruction effect that significantly reduces CSPE. The ADCV policy typically involves high antidumping and countervailing duties, raising the price of CPVP in target markets and increasing import costs. This diminishes the price competitiveness of CPVP. Additionally, countries implementing such policies create non-tariff barriers through legal and administrative measures, restricting market access and limiting sales opportunities for CPVP, further reducing their competitiveness and export volume.

Table 2
Descriptive statistics.

variable	N	mean	p50	sd	min	max	range
CSPE	2421	14.33	14.24	3.193	4.625	22.76	18.13
ADCV	2421	0.0900	0	0.286	0	1	1
lngdp	2421	24.43	24.35	2.215	18.70	30.69	11.99
lnpop	2421	15.78	16.02	1.969	9.777	21.05	11.27
lnind	2421	25.78	25.80	3.600	16.08	36.19	20.11
urb	2421	58.41	59.04	23.08	9.864	100	90.14
solar_ej	963	0.0170	0	0.0610	0	0.710	0.710
solar_twh	963	1.744	0.0440	6.403	0	75.14	75.14
solar_twh_net	963	1.726	0.0440	6.339	0	74.38	74.38
pvlplant	971	10.95	0	47.35	0	614	614
renewables_ej	968	0.128	0.0240	0.297	0	2.446	2.446
ren_power_ej	968	0.107	0.0180	0.247	0	2.304	2.304
ren_power_twh	968	10.48	1.781	24.27	0	232.8	232.8
ren_power_twh_net	968	10.21	1.729	23.73	0	228.4	228.4
electbyfuel_ren_power	350	22.15	8.069	36.83	0	232.8	232.8

Table 3
Baseline regression.

	(1)	(2)	(3)	(4)	(5)
	CSPE	CSPE	CSPE	CSPE	CSPE
ADCV	−1.7587***	−1.6453***	−1.5710***	−1.5488***	−1.5344***
	(0.1474)	(0.1475)	(0.1456)	(0.1460)	(0.1450)
lngdp		0.5768***	0.3511**	0.7494***	0.7039***
		(0.1510)	(0.1601)	(0.2119)	(0.2119)
lnpop			2.8170***	3.1014***	2.4420***
			(0.5981)	(0.6101)	(0.6157)
lnind				−0.6668***	−0.6612***
				(0.2450)	(0.2420)
urb					0.1084***
					(0.0275)
_cons	14.4868***	0.3828	−38.5508***	−35.5846***	−30.5465***
	(0.0302)	(3.6937)	(9.1133)	(9.1028)	(9.0269)
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
R^2	0.8336	0.8347	0.8365	0.8372	0.8385
N	2420	2420	2420	2420	2420

Standard errors in parentheses.
$^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2. Robustness tests

4.2.1. Parallel trend test and dynamic test

To ensure the validity of the staggered DID model, this study begins with a parallel trend test. This ensures that the ADCV effects estimated by the model have causal explanatory power by testing whether the treatment and control groups have the same trend in CSPE before the ADCV is implemented. Following the approach of Deschenes et al. (2017), the model is as follows:

$$CSPE_{it} = \beta_0 + \sum_{s=1}^{3} \beta_{pre_s} D_{pre_s} + \beta_{current} D_{current} + \sum_{s=1}^{8} \beta_{post_s} D_{post_s} + \sum \beta_i control_{it} + \lambda_i + \delta_t + \epsilon_{it} \quad (2)$$

Fig. 3 shows that before period 0, the regression coefficients are not significant, indicating that the treatment and control groups have consistent trends prior to ADCV implementation, thus satisfying the parallel trends assumption. At the same time, the coefficients after period 0 are significantly negative, indicating the negative impact of ADCV on CSPE. It is worth noting that the ADCV coefficient remains significantly negative in the seventh and eighth years following policy implementation, though to a lesser extent than in the third to sixth years. This could be attributed to the fact that most of the treatment group consists of EU countries, which imposed ADCV measures on China in 2012 but lifted tariffs on CPVP in 2018. Overall, the consistently negative and significant ADCV coefficient suggests that the policy has had a substantial and enduring impact on CSPE, clearly damaging trade.

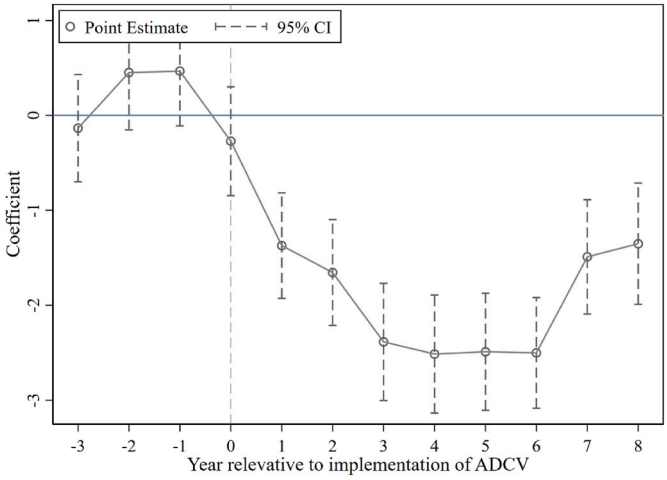


Fig. 3. Parallel trend test results.

Fig. 3 shows the results of the parallel trend test. In this specification, year −4 (pre_4) is set as the baseline period and is therefore omitted from the regression. The coefficients for the years prior to period 0 (i.e., pre_3, pre_2, pre_1) are statistically insignificant, suggesting that the

treatment and control groups followed similar trends before the implementation of ADCV measures, thus satisfying the parallel trends assumption. In contrast, the coefficients for the post-treatment years are significantly negative, indicating that ADCV measures have a substantial adverse impact on CSPE. Notably, the negative effect remains statistically significant even in the seventh and eighth years following policy implementation, although the magnitude of the effect decreases relative to years three through six. This pattern may be explained by the fact that most treated countries in our sample are EU members, which imposed ADCV measures on China in 2012 and lifted tariffs on CPVP in 2018. Overall, the results suggest that ADCV has a long-lasting and detrimental impact on trade, as reflected by the consistently negative and significant coefficients.

To complement the graphical evidence, we additionally report the results of an event-study regression in Table 4, which formally tests the parallel trends assumption. As shown, the coefficients for the pre-treatment periods (pre3 to pre1) are small and statistically insignificant (p-values >0.1), indicating no differential trends between the treatment and control groups before policy intervention. In contrast, the post-treatment coefficients (post1 to post8) are significantly negative at the 1 % level, with the most pronounced effects observed between post3 and post6. These findings provide robust empirical support for the validity of the difference-in-differences strategy and confirm that the parallel trends assumption holds in our study.

4.2.2. Placebo test

To validate the robustness of our model, we conduct a placebo test. This test helps rule out potential endogeneity issues and pseudo-regression results, ensuring that the estimated ADCV effects are genuine and not driven by unobserved factors or time trends. Following the methods of Chetty et al. (2009), we conducted a nonparametric placebo test by randomly sampling regions and times. The results, shown in Fig. 4, indicate that the random estimates are mostly centered around zero, with few reaching significance. In contrast, the estimated coefficient (−1.5344) from Table 4 lies in the lower tail of the distribution, indicating that the baseline regression passes the placebo test.

4.2.3. PSM-DID

To address potential selection bias and endogeneity issues, we further apply the PSM-DID method. This combined approach allows us to better account for unobserved heterogeneity and control for confounding factors that might distort the causal relationship between ADCV and CSPE. Specifically, we use yearly matching to ensure that the units are matched based on their characteristics in each year, accounting for temporal variations in the treatment assignment. Additionally, we employ a 1:2 nearest neighbor matching technique, where each treated unit is matched with two control units that have the closest propensity scores. Column (1) of Table 4 illustrates the ADCV coefficients remain significantly negative, confirming the robustness of the findings.

To address potential selection bias and endogeneity concerns, we employ the Propensity Score Matching combined with Difference-in-Differences (PSM-DID) approach. This method helps control for confounding variables and better identify the causal impact of ADCV measures on CSPE. Propensity scores are estimated using a logistic regression model based on pre-treatment covariates, including GDP, population, industrial value added, and urbanization rate. Year-by-year matching is conducted to account for temporal variations and ensure comparability within each period. The matching procedure adopts 1:2 nearest neighbor matching without replacement, with a caliper of 0.05 to restrict matches within a reasonable range of propensity scores. We enforce the common support condition to exclude unmatched observations and apply a tie-handling rule to address cases of identical scores during neighbor selection.

This matching procedure ensures that treated and control units are sufficiently similar in observed characteristics, which strengthens the credibility of the causal inference. Column (1) of Table 4 presents the regression results after applying the PSM-DID method. The estimated coefficient on ADCV remains significantly negative, consistent with our baseline findings, thereby reinforcing the robustness of the results.

Table 4
Event-study regression results: Dynamic treatment effects.

Time Relative to Treatment	Coefficient	p-value
pre3 (−3)	−0.1348	0.640
pre2 (−2)	0.4517	0.143
pre1 (−1)	0.4668	0.113
current (0)	−0.2724	0.352
post1 (+1)	−1.3722	0.000
post2 (+2)	−1.6560	0.000
post3 (+3)	−2.3866	0.000
post4 (+4)	−2.5146	0.000
post5 (+5)	−2.4907	0.000
post6 (+6)	−2.5025	0.000
post7 (+7)	−1.4913	0.000
post8 (+8)	−1.3529	0.000

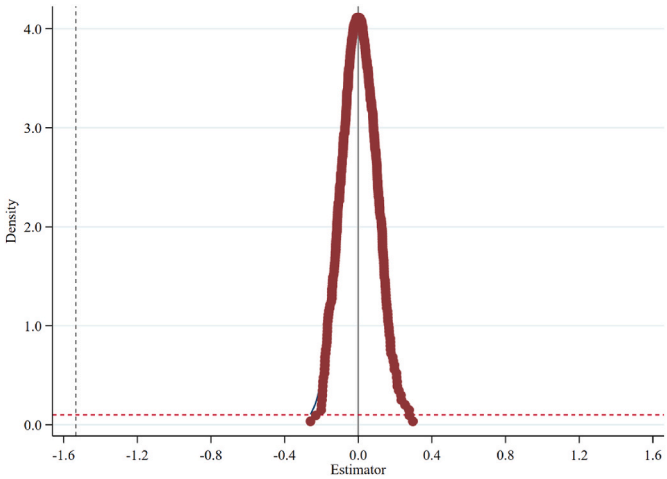


Fig. 4. Placebo test.

4.2.4. Data exclusion criteria

To further test the robustness of our results, we apply shrink-tailed 1 % and 5 % tests. These tests exclude the extreme 1 % or 5 % of observations from both ends of the sample distribution, reducing the influence of outliers on the estimation. This approach ensures that our regression results are not overly affected by anomalous data points, strengthening the robustness and reliability of the findings. As shown in columns (2) and (3) of Table 4, the ADCV coefficient remains significant.

4.2.5. Substitute the dependent variable test

To further validate the robustness of our results, we replace the original export trade value with two alternative measures: weight (the total weight of each shipment) and quantity (the number of units in each shipment) of CSPE. These indicators offer a more direct assessment of export volume, free from the influence of price fluctuations or other external factors. By using weight and quantity, we can better capture the actual scale of CSPE and confirm the robustness of our original regression findings, minimizing potential distortions from price changes. The results, as shown in Column 4 and Column 5, remain significant.

4.2.6. Country-level clustered standard errors

To further validate the robustness of our baseline findings, we re-estimate the main specification using standard errors clustered at the country level. The results, reported in Column (6) of Table 5, show that the coefficient on ADCV remains significantly negative and similar in magnitude (−1.5344). This finding reinforces our conclusion that anti-dumping and countervailing investigations significantly reduce

Table 5
Other robustness tests.

	(1)	(2)	(3)	(4)	(5)	(6)
	CSPE	CSPE	CSPE	lnweight	lnqty	CSPE
ADCV	−1.4271***	−1.5091***	−1.3934***	−1.7458***	−0.7018***	−1.5344***
	(-6.8321)	(0.1430)	(0.1325)	(0.2534)	(0.1818)	(0.2140)
_cons	−30.3233***	−30.3038***	−28.0828***	0.4302	−1.1e+02***	−30.5465*
	(-2.8069)	(8.9285)	(8.2842)	(18.0740)	(15.6463)	(15.5730)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.8557	0.8415	0.8465	0.7986	0.8347	0.8385
N	1435	2420	2420	1417	2056	2420

Standard errors in parentheses. Column (6) reports results using country-level clustered standard errors.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

China's photovoltaic export value and confirms the robustness of our results to alternative clustering levels.

4.2.7. Dynamic panel models

To further test the robustness of the results, we apply a Dynamic Panel Data Model (DPDM) using the system GMM estimator. This method addresses the endogeneity of the lagged dependent variable and unobserved heterogeneity by incorporating internal instruments, as proposed by Blundell and Bond (1998). It also controls for individual and time fixed effects, enhancing the credibility of the estimated policy effects. The results presented in Table 6 show that the coefficient of ADCV is significantly negative, suggesting that ADCV measures reduce CSPE, consistent with our baseline findings. The lagged dependent variable is also significant and positive, indicating path dependence in export performance.

The validity of the GMM estimates is assessed through several diagnostic tests. The Arellano–Bond test reports significant first-order autocorrelation [AR(1) $p < 0.001$], which is expected in differenced residuals and does not indicate model misspecification (Arellano and Bond, 1991). Crucially, the test for second-order autocorrelation [AR(2) $p = 0.207$] is insignificant, satisfying the key assumption of no second-order serial correlation in the error term. Furthermore, the Hansen J-test ($p = 0.183$) does not reject the null hypothesis of instrument validity, suggesting that the instruments used are appropriate and the model is not over-identified. These results confirm the robustness of our main conclusion: ADCV measures significantly hinder China's photovoltaic export, and the dynamic panel approach strengthens the causal inference.

4.2.8. Robustness check: inclusion of gravity variables

Considering that trade openness and bilateral distance are two important variables frequently used in gravity models, we conduct an additional robustness check by including open (trade openness) and dist (bilateral distance) as control variables in the baseline regression model.

Given that trade openness and bilateral distance are two variables frequently employed in gravity models, we perform an additional robustness check by incorporating open (trade openness, measured as the ratio of total trade to GDP) and dist (bilateral distance between China and its trading partners) into the baseline regression.

The regression results are reported in Table 7. As shown, the estimated coefficients for the policy variable ADCV remain significantly negative at the 1 % level across both specifications, confirming the robustness of our main findings. It is worth noting that dist is automatically omitted due to perfect collinearity with the fixed effects. This is expected, as bilateral distance is a time-invariant variable and is absorbed by the inclusion of both country and year fixed effects. Despite this, the stability of the ADCV coefficient suggests that our key results are not driven by omitted variable bias related to trade openness or bilateral distance.

5. Further analyses

5.1. Impact mechanism analysis: trade deflection and domestic sales

Through additional tariffs, the countervailing and anti-dumping policies of the EU, US, and other countries have sharply increased the

Table 6
GMM results.

	(1)
	CSPE
L.CSPE	0.3148***
	(0.0658)
ADCV	−0.7348***
	(0.2207)
AR(1)	(0.000)
AR(2)	(0.207)
Hansenp	(0.183)
_cons	−7.5349***
	(1.2440)
Controls	Yes
Country FE	Yes
Year FE	Yes
N	2183

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table 7
Robustness test with trade openness and distance.

	(1)	(2)
	CSPE	CSPE
ADCV	−1.5492***	−1.6181***
	(0.1450)	(0.1476)
lngdp	0.7047***	0.7073***
	(0.2222)	(0.2246)
lnpop	2.4645***	2.3066***
	(0.6175)	(0.6324)
lnind	−0.6459**	−0.6315**
	(0.2508)	(0.2591)
urb	0.1079***	0.1072***
	(0.0275)	(0.0276)
open	0.0023	0.0020
	(0.0021)	(0.0022)
dist		0.0000
		(omitted)
_cons	−31.4168***	−29.2681***
	(9.0817)	(9.3698)
Country FE	Yes	Yes
Year FE	Yes	Yes
R^2	0.8385	0.8403
N	2402	2336

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

local prices of China's photovoltaic products, eroding their price competitiveness and hindering exports to these markets. China's photovoltaic industry has had to explore new markets to mitigate the negative impact of these trade barriers. From a foreign demand perspective, as exports to the EU and US are restricted, the industry actively seeks new export destinations, leading to trade deflection toward third-party countries. From a domestic demand perspective, export disruptions may shift some production to the domestic market, with the establishment of photovoltaic power facilities absorbing part of the excess capacity and reducing the adverse effects of trade disruption.

Following trade disruptions in the Chinese PV industry, exports have shifted to other third-party countries. These new export markets are usually characterized by higher PV market demand, better PV infrastructure, and an ongoing energy transition process. The mechanisms behind the trade deflection mainly include the reconfiguration of market demand, the support of infrastructure development, and the driving force of the policy and economic environment. Therefore, in subsections 5.1.1 to 5.1.3 of this paper, the trade deflection effect will be explored through an empirical study of PV market demand, PV infrastructure development, and the energy transition process. In subsection 5.1.4, this paper will explore China's response to the adverse impact of the double-reverse policy through the construction of domestic PV power plants, thus realizing the domestic sales of some PV products. Fig. 5 illustrates the mechanism diagram.

To empirically investigate the deflection effect of anti-dumping and countervailing (ADCV) measures on China's photovoltaic (PV) exports, this paper adopts a unified interaction regression framework. The goal is to examine whether China's exports are redirected to countries not imposing such measures, and whether specific country-level characteristics moderate this redirection. The general specification is as follows:

$$CSPE_{it} = \beta_0 + \beta_1 Post_NonADCV_{it} + \beta_2 Post_NonADCV_{it} \times M_{it} + \beta_3 M_{it} + \sum \beta_i control_{it} + \lambda_i + \delta_t + \epsilon_{it} \quad (4)$$

In Equation (4), $CSPE_{it}$ denotes the export value of Chinese PV products to country i at time t. The key variable $_NonADCV_{it} = Post_t \times NonADCV_i$, where $Post_t$ is a dummy variable indicating the period after the imposition of ADCV measures, and $NonADCV_i$ is a dummy equal to 1 for countries that did not impose ADCV measures on China. M_{it} stands for different moderating variables, representing country-level characteristics that may influence the strength of the deflection effect.

A significantly positive β_1 indicates the existence of a trade deflection effect, meaning that following the imposition of ADCV measures by countries such as the US and the EU, China's PV exports shifted toward unaffected third countries. In addition, β_2 is the coefficient of primary interest in this analysis. A significantly positive β_2, the coefficient on the interaction term $Post_NonADCV_{it} \times M_{it}$, suggests that the moderating variable M strengthens the deflection effect, implying that Chinese PV exports are more likely to be redirected to third countries with such characteristics when facing ADCV measures.

To systematically explore how China's PV exports are deflected in response to ADCV measures, we select a series of moderation variables that capture key structural and developmental features of potential importing countries. Specifically, these variables represent differences in solar market demand, infrastructure capacity, and the depth of renewable energy transition, which may shape the redirection pathways of trade flows.

First, regarding the trade deflection mechanism, market demand in destination countries plays a central role in determining where Chinese PV exports are redirected. According to the heterogeneous firm trade theory, export flows are more likely to be diverted toward countries with substantial and growing market potential (Helpman et al., 2008). In the PV sector, such demand is captured through indicators reflecting solar energy consumption and electricity generation. We include solar_ej (solar energy consumption in exajoules), solar_twh (total solar electricity generation), and solar_twh_net (net solar electricity generation after losses) to measure both the scale and depth of solar market development. These indicators jointly reflect a country's PV market demand, which influence its attractiveness as a substitute destination. Previous studies have also suggested that countries with high solar electricity generation are more likely to become substitute destinations for PV exports following trade restrictions in traditional markets (Gereffi et al., 2005; IEA, 2024; IRENA, 2023).

Second, beyond market demand, infrastructure capacity in importing countries determines their ability to utilize and integrate imported solar equipment. This relates to the concept of absorptive capacity, which emphasizes not only technological openness but also the presence of complementary assets required for effective deployment (Manu et al., 2022; Ibrahim et al., 2018). In the PV industry, infrastructure readiness is best proxied by the deployment of large-scale solar power plants. We use pvlplant, the number of utility-scale PV power plants (≥20 MW), to reflect a country's capacity to absorb redirected exports through existing grid and institutional structures. As noted by IEA-PVPS (2024), infrastructure development often precedes policy scaling and broader market uptake, making it a critical moderator of trade diversion effects.

Third, we consider the broader renewable energy transition as a key factor influencing trade deflection. Countries undergoing faster transitions to renewable energy tend to exhibit supportive policy environments, higher levels of clean energy investment, and stronger technology adoption capabilities (Sovacool, 2016; Geels et al., 2017). These dynamics can amplify demand for imported PV products. To capture this mechanism, we include five indicators: renewables_ej (total renewable energy consumption), ren_power_ej (renewable electricity consumption), ren_power_twh (renewable electricity generation), ren_power_twh_net (net renewable electricity generation), and elecbyfuel_ren_power (the share of electricity generated from renewable sources). These variables reflect the pace, scale, and integration level of the energy transition process, which jointly determine the likelihood of a country becoming a redirected destination for PV exports. This aligns with prior research, which has also shown that countries undergoing rapid renewable adoption tend to exhibit stronger import demand for clean energy products and solutions (Xu and Sharma, 2022; IRENA, 2023).

5.1.1. Trade deflection: photovoltaic market demand

The results in columns (1)–(3) of Table 8 show that the estimated coefficients of Post_NonADCV × solar_ej, Post_NonADCV × solar_twh, and Post_NonADCV × solar_twh_net are all significantly positive at the 1 % level, suggesting that PV market demand amplifies the trade

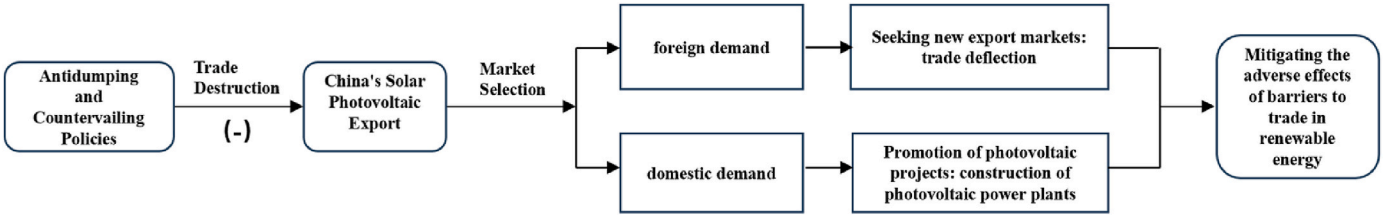


Fig. 5. Mechanism diagram.

Table 8
Moderating effects of photovoltaic market demand.

	(1)	(2)	(3)
	CSPE	CSPE	CSPE
Post_NonADCV	1.3502***	1.3525***	1.3525***
	(0.2059)	(0.2057)	(0.2057)
Post_NonADCV × solar_ej	9.1105***		
	(1.6404)		
Post_NonADCV × solar_twh		0.0863***	
		(0.0155)	
Post_NonADCV × solar_twh_net			0.0871***
			(0.0157)
_cons	−51.9339***	−51.8269***	−51.8269***
	(15.2564)	(15.2529)	(15.2529)
Controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
R^2	0.8512	0.8512	0.8512
N	963	963	963

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table 9
Moderating effects of photovoltaic infrastructure implementation.

	(1)
	CSPE
Post_NonADCV	2.1271***
	(0.2434)
Post_NonADCV × pvlplant	0.0088**
	(0.0044)
_cons	−59.6818***
	(18.3987)
Controls	Yes
Country FE	Yes
Year FE	Yes
R^2	0.8336
N	971

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

deflection effect of the ADCV policy on CSPE. In other words, China's solar exports tend to be redirected to countries with high PV power generation capacity and strong market demand.

First, countries with higher PV power generation capacity usually invest more in the PV field, including technology R&D and policy support. With these countries' rapid development of PV power generation projects, their investment needs and market demand also increase significantly. Second, expanding countries with large PV power generation requires many PV products, such as PV modules. At the same time, domestic production capacity and technology levels often make it challenging to meet the growing demand quickly. To ensure the successful implementation of PV power generation projects and meet the target output, the country will increase the import of efficient and high-quality PV products from countries with advanced technology and strong production capacity. Finally, China's solar photovoltaic sector, a global leader with strong manufacturing capacity, advanced technology, and cost advantages, will seek new markets after encountering the double-reverse, and exports will be deflected towards these countries with large PV power generation and strong demand in the PV market. Empirical evidence also suggests that Chinese PV product exports to countries with large PV power generation have increased significantly after the EU and the US implemented PV double-reverses against China. For example, as of 2020, Japan has more than 70 GW of installed PV capacity and generates about 85 TW h (TWh) of PV power per year, India has about 40 GW of installed PV capacity and generates about 50 TW h (TWh) of PV power per year. The Indian government launched the National Solar Mission with a target of 100 GW of installed solar capacity by 2022. Despite India's implementation of ADCV on CPVP in 2018, the vast market demand still attracts many Chinese PV products.

5.1.2. Trade deflection: photovoltaic infrastructure implementation

Table 9 show that the estimated coefficients of ADCV × pvlplant is significantly positive, suggesting that China's PV export tendency is deflected towards countries with a large number of PV power plants. After China's PV products were subjected to ADCV measures by Europe and the United States, its export strategy was significantly adjusted and gradually shifted to countries with well-developed PV infrastructures and many PV power plants.

These countries usually invest heavily in the PV energy sector, covering technology research and development, policy incentives, and project construction to accelerate the growth of their renewable energy sector. For example, as of 2020, the UAE has installed a cumulative total of about 2.5 GW of installed PV capacity, mainly through large-scale PV power plants. The Noor Abu Dhabi project in the UAE is one of the largest single PV plants in the world, with an installed capacity of 1.17 GW. The Benban Solar Park in Egypt is one of the world's most significant solar power projects, with a total installed capacity of 1.65 GW. As the construction of PV power plants continues to expand, the demand for high-quality PV modules in these countries has increased dramatically. China's PV industry occupies an crucial position in the global market due to its leading manufacturing capability, advanced technology, and cost advantage. After implementing ADCV measures in Europe and the United States, the photovoltaic sector in China has gradually shifted to transferring new markets, especially those countries with rapid growth in PV infrastructure and power plant construction. By exporting to countries with perfect PV infrastructure and a high number of PV power plants, Chinese PV products not only found a new market outlet but also promoted the further development of the global PV industry.

5.1.3. Trade deflection: energy transition process

The interaction terms ADCV × renewables_ej, ADCV × ren_power_ej, ADCV × ren_power_twh, ADCV × ren_power_twh_net, and ADCV × electbyfuel_ren_power in Table 10 all have significantly positive coefficients. This indicates that the moderating variables amplify the trade deflection effect of the ADCV policy on CSPE, as China's PV exports tend to be increasingly redirected toward countries undergoing rapid energy transitions.

Firstly, such countries typically offer strong policy support, including subsidies and tax incentives, to boost renewable energy development. For instance, India's national solar program, Japan's feed-in tariff system, and Australia's renewable energy target have significantly increased PV market demand. These policy-driven markets attract China's PV industry, seeking to capitalize on emerging opportunities. Secondly, the double-reverse measures in Europe and the US have restricted Chinese PV sales, prompting the industry to explore new markets. Countries with fast energy transitions, characterized by high PV demand and rapid market growth, offer ideal alternatives, helping China diversify its market risks and enhance competitiveness. Lastly, China has actively pursued international cooperation with these countries, signing agreements and launching joint PV projects. For example, China's collaboration with India has expanded, with Chinese firms establishing production bases and sales networks, deepening market penetration. This strategic cooperation has bolstered China's PV industry's global competitiveness while promoting economic and technological exchanges.

5.1.4. Domestic sales: construction of domestic PV power plants

After the EU and the US implemented ADCV measures on CPVP, China's PV industry underwent significant adjustments, shifting to domestic sales becoming a key strategy. In addition to seeking third-party export markets, the domestic market gained prominence. Domestic PV power plant construction and power generation surged during the double-reverse period.

Table 10
Moderating effects of energy transition process.

	(1)	(2)	(3)	(4)	(5)
	CSPE	CSPE	CSPE	CSPE	CSPE
Post_NonADCV	1.2579***	1.2342***	1.2320***	1.2312***	1.0128***
	(0.2109)	(0.2138)	(0.2139)	(0.2139)	(0.3827)
Post_NonADCV × renewables_ej	1.6084***				
	(0.3396)				
Post_NonADCV × ren_power_ej		2.1813***			
		(0.4847)			
Post_NonADCV × ren_power_twh			0.0222***		
			(0.0050)		
Post_NonADCV × ren_power_twh_net				0.0229***	
				(0.0051)	
Post_NonADCV × electbyfuel_ren_power					0.0208***
					(0.0058)
_cons	−59.5847***	−60.8609***	−60.7468***	−60.7773***	−42.6603
	(15.4047)	(15.5132)	(15.5038)	(15.5036)	(38.3704)
Controls	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
R^2	0.8499	0.8499	0.8500	0.8500	0.7955
N	968	968	968	968	350

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

As shown in Fig. 6, the number of large-scale PV plants increased from 91 in 2012 to 3185 in 2020, and small to medium-sized plants rose from 13 to 195. Similarly, Fig. 7 illustrates the dramatic growth in China's PV power generation during this period.

Government support largely drove this expansion through initiatives like the "frontrunner" program and policies promoting PV power plant construction. By integrating upstream and downstream resources and focusing on the Western region, China transformed its market structure and developed a new industrial framework. This strategic shift helped mitigate the negative impacts of the double-reverse policy by encouraging large-scale domestic PV development.

5.2. Heterogeneity analyses

5.2.1. Heterogeneity in the strength of antidumping and countervailing policies

The negative impact of ADCV on CSPE varies depending on the country implementing them. The paper classifies the ADCV countries into the EU, the U.S., and others. As shown in Fig. 8, all ADCVs significantly hinder China's PV exports, with the EU and U.S. having a more pronounced trade-damaging effect than other countries.

This disparity stems from several factors. First, the EU and U.S. impose stricter antidumping measures with higher tax rates, broader coverage, and more vigorous enforcement, creating substantial trade barriers for China's solar energy sector. Second, the EU and the U.S. have more advanced PV industries, and their policies aim to protect domestic markets from global competition, particularly from China, further intensifying market entry challenges. In contrast, countries like Canada, India, and Turkey, with less developed PV industries,

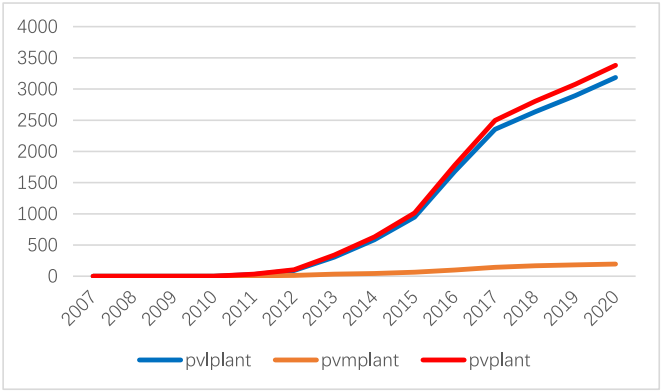


Fig. 6. Number of PV plants built in China.

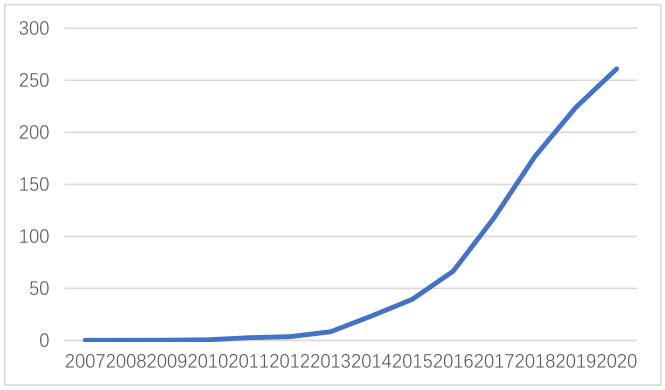


Fig. 7. China's photovoltaic power generation(twh).

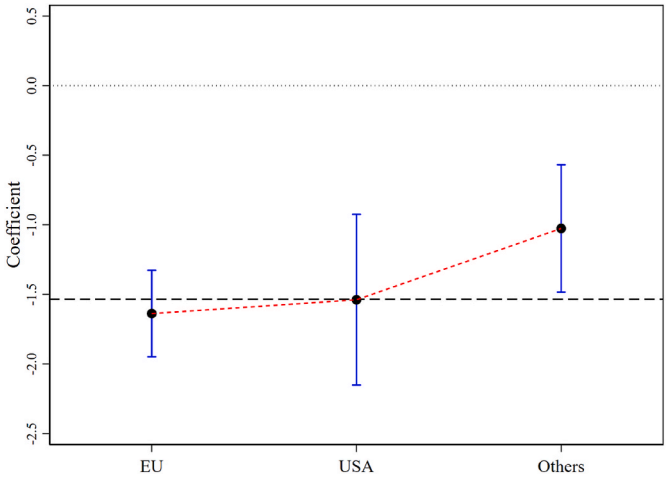


Fig. 8. Heterogeneous results on the strength of ADCV policy.

implement milder ADCVs primarily to protect nascent sectors. Lastly, the EU and U.S., as major global economies, exert more significant influence in international trade, making their ADCVs more impactful and creating higher trade barriers for China's PV exports.

5.2.2. Heterogeneity analysis of high-energy investment regions

The impact of ADCV on CSPE varies depending on the level of energy investment in the importing countries. Fig. 9 shows that the ADCV coefficient is significantly negative for high-energy investment countries, while it is not significant for low-energy investment countries. It indicates that the negative impact of the ADCV policy is stronger in markets with higher energy investments.

Several factors explain this difference. First, high-energy investment countries strongly demand renewable energy and are critical markets for Chinese PV products. Chinese companies rely more on these markets, so the dual-reverse policy presents a more significant challenge to their exports. Second, these countries offer higher returns on investment, attracting numerous international investors. The policy weakens the competitiveness of Chinese PV products, reducing investor demand and exacerbating the export suppression effect. Lastly, the competition in these markets is intense. When high-energy investment countries impose ADCV measures, Chinese PV firms lose their price advantage, leading to a rapid decline in market share. As a result, the inhibitory effect of the dual-reverse policy on Chinese PV exports is more pronounced in these markets.

5.2.3. Heterogeneity analysis of high-energy dependent regions

The effect of ADCV on CSPE may vary across regions with different degrees of energy dependence. Fig. 10 shows that the ADCV policy significantly inhibits CSPE in high energy-dependent countries, while the effect is insignificant in low energy-dependent countries.

First, countries with high energy dependence are highly dependent on imported energy and usually pay more attention to protecting their energy industries, including the PV industry, to reduce their dependence on external energy sources. Through strict double-anti policies, these countries try to protect local producers from the impact of imported products. As a result, the double-reverse policy against Chinese PV products will be more stringent, leading to more excellent trade-disrupting effects. Second, countries with high energy dependence usually have a high demand for renewable energy, large market sizes, and a large share of Chinese PV products in these markets. When these countries implement ADCV policy, the impact on Chinese PV firms is more significant, leading to more excellent trade-damaging effects. Finally, to ensure economic and energy security, countries with high energy dependence will implement ADCV more strictly to prevent many low-priced imports from entering the market and to protect local enterprises and market stability. This strict policy enforcement increases the difficulty for Chinese PV products to enter these markets, leading to more excellent trade-damaging effects.

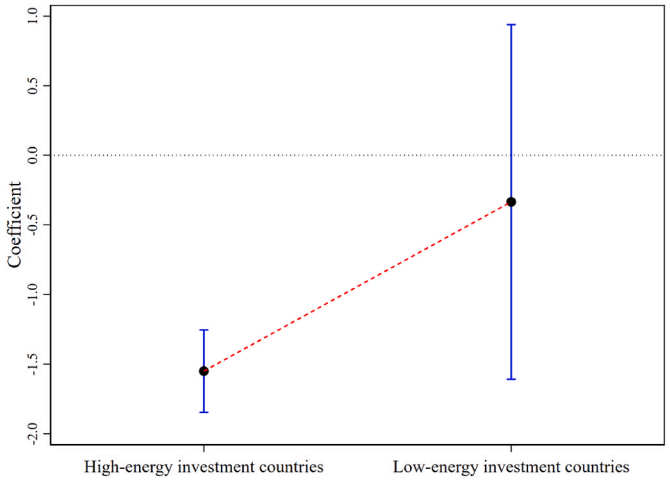


Fig. 9. Heterogeneity analysis of High-energy investment regions.

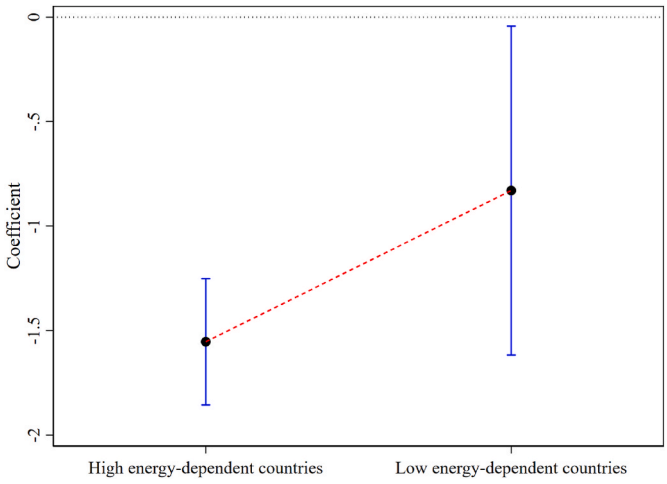


Fig. 10. Heterogeneity analysis of High-energy dependent regions.

6. Conclusion and policy implications

6.1. Conclusions

Amid the worldwide shift toward sustainable energy and the increasing prevalence of international trade protectionism, examining the impact of ADCV on CSPE has become critical. Using panel data on PV trade between China and other countries from 2007 to 2020, this research examines the negative impact of ADCV on CSPE and their shift to third-party markets in response. The main findings are as follows:

First, anti-dumping and countervailing policies have a trade-destroying effect, significantly reducing China's PV exports. These conclusions remain robust after a series of tests. Second, when China's PV exports are restricted by trade barriers in countries like the EU and the US, they are deflected to third-party markets with high PV demand, substantial infrastructure development, and rapid energy transitions. Additionally, some production capacity is redirected to the domestic market to mitigate the adverse effects of trade disruption. Third, heterogeneity analysis shows that the impact of these policies varies significantly across countries, depending on the strength of policy enforcement, levels of energy investment, and energy dependence. These findings offer valuable insights for China in addressing renewable energy trade barriers, optimizing the PV industry's overseas market strategy, and promoting the global energy transition.

6.2. Policy implications

First, strengthen diversified market development and optimize the layout of overseas market capacity. China's photovoltaic industry should make more significant efforts to develop third-party countries outside of Europe and the United States market, especially countries with sizeable photovoltaic market demand, construction of photovoltaic infrastructure, and energy transition processes. It will help reduce the export losses brought about by the ADCV policy in Europe and the United States and enhance the competitiveness of CPVP worldwide. China's photovoltaic industry needs to further optimize and accelerate its overseas development, with increased investment in key markets to establish factories, thereby enhancing internationalization and mitigating the negative impact of ADCV.

Secondly, photovoltaic technology and cost competitiveness should be consolidated, and the domestic market development should be

promoted. In the face of international trade barriers, China should increase support for the domestic PV market, rationalize the layout of domestic PV production capacity, and promote the construction of domestic PV power plants in an orderly manner. Through policy guidance, financial subsidies, technological innovation support, and other initiatives, promote the advancement of domestic photovoltaic energy projects to enhance the absorption capacity of the domestic market for PV products to alleviate the impact of external market uncertainty on CSPE.

Finally, flexible response strategies should be developed according to the market demand and policy variations across regions. China should adopt a flexible response strategy for different countries' policy implementation strengths, energy investment levels, and energy dependence differences. For example, countries with robust policy implementation can actively respond through legal proceedings and international negotiations; for countries with high levels of energy investment, it can strengthen cooperation and exchanges and jointly promote energy transformation projects; for countries with high energy dependence, it can enhance the competitiveness of photovoltaic products in the local market by providing technical support and product advantages. By adopting differentiated strategies, China can mitigate the harmful consequences of ADCV, boosting the global competitiveness of China's photovoltaic sector and enhancing its resilience to trade barriers.

CRediT authorship contribution statement

Boqiang Lin: Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Conceptualization. **Yida Liu:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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